Responsive dribbling for Touchline

EA SPORTS FC 26 and FC 27 research and architecture review

Prepared for Touchline • 20 October 2026

Research and design only. No Touchline implementation or repository changes were made.

1 Executive conclusion

Touchline should give its simulation direct authority over responsive player movement and conditional ball contacts, then make the body adapt to those decisions. Exact support-foot choreography should not be a prerequisite for ordinary dribbling.

REASONABLE INFERENCE. The evidence points to a coordinated combination of early response, short action entry and recovery, context-sensitive motion selection, and animation that can adapt during execution. EA has documented this design direction through both contemporary animation interviews and historical FIFA technical material. There is no public specification of one complete FC 26 dribbling algorithm. [S3, S10, S12]

For Touchline, the decisive distinction is between acknowledging a request, reaching the next legitimate contact, and completing the requested movement. A player can begin braking, opening the body, or adjusting a step immediately. The ball can still travel under simulation until a foot genuinely reaches it. The completed turn can take longer without the controls feeling ignored.

The proposed change is to make ordinary carrying a continuously revisable control activity. Most analog input should modify movement and the next uncommitted touch. It should not purchase an entire named move that must finish before control returns. Preserve brief, meaningful commitments around contacts and genuine physical constraints, then give presentation substantial freedom over how the feet and body get there.

Evidence boundary. The supplied brief establishes reliable retention at Ball Control 99, Dribbling 99 and Agility 99, followed by poorer response and visuals when detailed body mechanics were enforced. No PLAYTEST-1.2 build, source code or raw trace was supplied or inspected. The diagnosis below is therefore a testable explanation of the described architecture, not a finding from a code audit. The A/B proposal explicitly uses the existing build as the comparison baseline. [B]

Evidence labels used throughout

Label	Meaning
OFFICIALLY CONFIRMED	An EA publication or identifiable developer statement describes the feature. This establishes the disclosure, not flawless behaviour in every shipped context.
OBSERVED	A visible property of footage actually inspected. It does not identify the hidden implementation.
REASONABLE INFERENCE	A design explanation supported by evidence but not disclosed as proprietary architecture.
SPECULATION	An unverified possibility. It is not used as a factual premise.
RECOMMENDATION	An original Touchline design or experimental target, not an EA measurement.

2 Confirmed EA design information

2.1 What the FC 26 gameplay notes actually confirm

OFFICIALLY CONFIRMED. The July 2025 deep dive supports all the principal claims identified in the brief. Its Dynamic Dribbling features are marked for newer consoles and PC. [S1]

Area	Published change
Touch cadence	Retuned intervals and ball proximity; presets differ.
Foot choice	Revised preference logic and branching.
Lock-in	Quicker transitions with less animation trapping.
Effort Touches	Contextual use during drastic dribble turns.
Braking	Faster deceleration across players.
Selection	Better alignment with current movement speed.
Coverage	Relaxed requirements permit earlier animation starts.
Consistency	Reconciled contact timing, trajectories, poses and exit speed.

EA gives no numerical interval, angle threshold, branch window or root-motion formula in this disclosure. In particular, current speed is confirmed; a complete selection formula combining requested speed and incoming stride is not. [S1]

2.2 The documented cause of some animation delay

OFFICIALLY CONFIRMED. EA's separate responsiveness investigation identifies input processing and animation selection as causes of delay even offline. Selection considers ball distance, input angle and player speed. Sparse animation coverage could cause a longer action or a wait while the player was repositioned. EA also describes selection bugs and expanding coverage. [S2]

The input improvement quantified there is one frame in some first-time pass and shot situations. It is not a measurement of dribbling latency. A later launch note describes an approximately 12 ms improvement in average input round-trip latency through client/server synchronisation. That number is a reduction in a network-related measure, not total stick-to-screen time. [S2, S8]

2.3 Faster animation still needs compatible transitions

OFFICIALLY CONFIRMED THROUGH A DEVELOPER INTERVIEW. Senior animator Paul Baruzzi explains that response targets require abbreviated wind-ups and earlier exits, while compatible-pose thresholds prevent fast transitions looking messy. The interview prints the terms PTM and Matchmaking without defining their algorithms. It does not establish a universal motion-matching implementation for FC dribbling. [S3]

A Gamescom gameplay interview also describes removing or trimming motions that looked acceptable but responded too slowly. Its speakers are not clearly named in the article, so it is corroboration rather than the sole foundation of this report. [S5]

RECOMMENDATION. Relax exact anatomical entry conditions, but retain visual compatibility. A responsive system needs a usable transition into the new movement, not a fresh full-body animation restart on every input sample.

2.4 Elite dribbling is more than a retention rating

OFFICIALLY CONFIRMED HISTORICALLY. FIFA 23 describes ordinary left-stick Technical Dribbling separately from modifier-based dribble systems. It links touch precision and ball proximity to Dribbling and Ball Control; animation execution and transition speed also involve Agility, Reactions and Balance. Active Touch determines approaches to the ball, with ML-Flow operating between contacts. These are FIFA 23 disclosures, not verified FC 26 formulas. [S12]

In a named interview, lead gameplay producer Paul Parsons describes FC 26's dribbling work as coordination across input, animation and positioning. The accompanying report describes more contacts and quicker movement, with elite dribblers benefiting from faster animation. Treat the writer's praise of the feel as a hands-on impression, not a latency benchmark. [S4]

2.5 Competitive and Authentic are explicit design choices

OFFICIALLY CONFIRMED. Competitive favours quicker transitions, tighter carrying and faster skills. Authentic favours slower carrying and more realistic transitions. General fundamental improvements apply to both. [S1]

Parsons gives a concrete recovery example: the Authentic presentation can include an appeal to the referee after entanglement; Competitive prioritises returning the player to action. This is a choice about the time occupied by behaviour, not simply a universal animation-speed multiplier. [S4]

FC 27's Authentic Gameplay 2.0 adds weight to touches, turns and sprints while retaining responsiveness as an objective. Its weather and contact descriptions include lost footing and stumbles. EA does not publish corresponding inertia coefficients. [S7]

RECOMMENDATION. Touchline can choose prompt control and legible technical execution while retaining its own match tempo, fatigue, tactical decision-making and physical-ball rules. A management simulation does not need to import every Competitive balance choice to borrow its response principles.

2.6 FC 27 provides a directly relevant continuation

OFFICIALLY CONFIRMED. The July 2026 FC 27 deep dive describes off-balance dribbling for players qualifying through Balance, Agility and Dribbling. After certain bumps or challenges, the player may visibly stumble while remaining controllable and taking a touch in the chosen direction. The feature is listed for PS5, Xbox Series X|S, PC and Switch 2. [S6]

REASONABLE INFERENCE. This is strong evidence for permitting useful control during an imperfect body state. It does not prove a simulated balance controller, a specific contact scheduler, or immunity to valid tackles.

The same disclosure narrows some PlayStyle advantages, including dribble-animation and acceleration boosts. It reinforces the need to record card, attributes and PlayStyles when comparing elite gameplay. A visually similar player is not necessarily an equivalent test subject. [S6]

2.7 Direct evidence of tolerated visual imperfection

OFFICIALLY CONFIRMED THROUGH A DEVELOPER INTERVIEW. Gameplay director Kantcho Doskov acknowledges accepting clipping when resolving well-timed tackles. The surrounding report explicitly discusses a tackling leg visually passing through an opponent. This is a documented gameplay-versus-visuals compromise in tackling, not proof of a particular dribble-foot correction. [S21]

FC 25 separately describes rain/snow pass-slip animations whose presence does not alter the pass or shot result. It also moved the start of Let Ball Run to the request rather than waiting for ball arrival. These are useful precedents for cosmetic imperfection and early visual acknowledgement; neither proves that the whole EA engine shares Touchline's authority boundary. [S14]

2.8 The technical history supports adaptive authored motion

OFFICIALLY CONFIRMED HISTORICALLY. EA's 2010 GDC session describes procedural alteration of FIFA turn animations and fitting actions to a place and time. The original slides explicitly cover contact IK and retiming to accommodate incoming speed and touch timing. The slide titled Interruptions provides no textual specification of interrupt windows. [S10]

FIFA 22's notes describe runtime generation of ball-approach motion, including stride, cadence and transition adjustments. They also document faster dribble deceleration for a qualifying attribute combination. Those historical eligibility rules should not be transferred to FC 26 or Touchline. [S11]

EA's 2021 interaction presentation discusses predicted interception locations and motion-matched interactions using FIFA and Madden examples. Its 2023 presentation

confirms broader EA use of motion matching. These support a family of techniques, not the claim that all FC 26 left-stick motion uses one particular search algorithm. [S15, S16]

2.9 What remains undisclosed

SPECULATION IF ASSERTED AS EA FACT. No source examined supplies FC 26's controller polling rate, simulation frequency, deadzone/filter parameters, angle thresholds, analog-request queue policy, contact commitment horizon, exact pose tolerances, full root authority model or between-touch ball constraints. Nor does it establish how often a particular retail elite player can branch under a rapid 0° → 90° → 0° → −90° input trace.

Those gaps matter. The recommendations that follow are independently justified design proposals; they are not a reconstruction of proprietary EA code.

3 Observations from gameplay

3.1 Material inspected and limits

OBSERVED. I inspected extracted sequences from EA's FC 26 Dynamic Dribbling, Pro Agility and new-skill demonstration videos. These are edited demonstrations, including a testbed with simplified player models. Their raw input timestamps, tuning preset and complete attributes are not provided. The Dynamic Dribbling name bar shows MBAPP at 22.5 seconds; that identifies the testbed label, not a verified retail Mbappé card. [V1, V2, V3]

The inspected Dynamic Dribbling material extends through approximately 26 seconds. Video timestamps locate the examples; they are not measurements of simulation time or stick latency. A camera crossfade near 22.75 seconds must not be mistaken for an animation blend or pose snap. [V1]

Clip and location	Visible property	What it does not establish
Dynamic Dribbling, about 22.9-24.4 s	Short adjustments around the ball; body and ball do not maintain one rigid alignment.	The exact surface/instant of every collision.
Dynamic Dribbling, about 24.4-25.0 s	A compact redirect develops into an exit with changing stride length.	A particular named move or internal Effort Touch tag.
Pro Agility, about 8-12 s	Braking, torso reorientation and changes of movement line occur through compressed stepping.	Raw stick timing or an inertia formula.
New Skill Moves, about 0-3 and 6-13 s	Labelled techniques have conspicuous choreography and, in some cases, lifted ball travel.	That any similar-looking ordinary cut is the same action.

The Pro Agility and skill examples are separate clips. Their appearance supports comparisons of movement vocabulary, not a controlled speed comparison between systems. [V2, V3]

3.2 What the motion vocabulary suggests

REASONABLE INFERENCE FROM THE OBSERVATIONS. Compact touches can remain readable without making every support transition a separate event. At the normal gameplay camera, the most useful cues are the changed ball line, the player's preparation for it, and a coherent exit. Giving equal emphasis to every plant, pelvis transfer and recovery can obscure that larger action.

A Messi-like target should therefore be described in terms of small corrections, economical motion and quick exits. A permanently crouched pose is insufficient. Preferred-foot identity can appear in the side from which the ball is carried and the shape of a touch, while a reachable alternative foot prevents unnecessary waiting. These are proposed visual goals, not isolated findings about Messi's current retail animation set.

3.3 Ball freedom and visual cheating require careful language

OBSERVED LIMIT. The footage shows changing separation and alignment; it cannot distinguish a free physics trajectory from one guided by hidden animation or gameplay constraints. I did not derive metres, contacts per second or foot-slip distances from these edited views. Camera motion and the simplified models make such numerical claims unreliable.

Shortened or rotating support steps are visible at the level of overall choreography. Exact world-space foot sliding, specific root-warp amounts, and pose snapping inside uninterrupted retail gameplay were not established. The direct evidence for intentionally tolerated clipping comes from the developer interview in section 2.7, not from treating ambiguous frames as proof.

A retail elite-dribbler video and the official FC 27 off-balance clip were located, but their video media could not be inspected in this session. Their titles, thumbnails and written descriptions are not counted as observations. The FC 27 findings in this report are supported by the published text. [S6]

3.4 Feel is sensitive to build and mode

OFFICIALLY CONFIRMED. EA acknowledged an unintended slowdown of Controlled Sprint in FC 26 and a subsequent live-tuning reversal. Any external benchmark therefore needs its build date and tuning version. [S9]

REPORTED HANDS-ON IMPRESSION. GamesRadar describes very small directional corrections, while the Epic hands-on report describes quick direction changes. These corroborate the perceived target. Neither provides synchronised stick, simulation and presentation traces. [S4, S21]

4 Inferences about likely architecture

4.1 Input response has several different clocks

REASONABLE INFERENCE. It is useful to distinguish at least four delays: receiving the input, visibly acknowledging it, making a relevant contact, and establishing the new movement line. These events need not share a gate.

Consider a sharp request while the ball is travelling between contacts. The player can immediately reduce forward velocity, rotate the upper body toward the intended exit and shorten the next step. The ball continues on its existing trajectory until a legitimate contact. After the redirect, the player accelerates onto the new line. An observer perceives a continuous response because the preparation begins before the outcome is complete.

If the whole response waits for the next clean support configuration, the user gets a period of silence followed by hurried correction. If the ball changes direction immediately without a contact, control may be fast but its physical explanation breaks. Touchline can avoid both by separating truthful preparation from contact execution.

4.2 A branch is not necessarily a complete animation cancellation

The evidence is consistent with several levels of adaptation: modify parameters of the current motion, enter another motion at a compatible point, abbreviate recovery, or select a different action family when the situation changes substantially. Public disclosures do not identify exactly which mechanism FC uses in every case.

For Touchline, a minor analog correction should usually alter direction, step placement or the next touch target while retaining the useful part of the current motion. An emergency reversal may need a different contact family. The entire character need not freeze merely because the contacting foot is briefly committed.

Pose matching and motion matching also solve different scopes of problem. Pose compatibility can be used within an ordinary transition graph. A broader motion search may choose a fragment using pose and desired motion information. Neither automatically gives good contact timing. A large library can still respond poorly if selection favours long actions or requires exact entry states.

4.3 Coverage has two complementary solutions

Coverage means having suitable motion for the relevant combination of situation, direction, speed and contact context. There are two distinct ways to improve the experience: add missing motion, and permit existing motion to fit a broader set of starting states.

RECOMMENDATION. Use both. More clips alone can multiply maintenance without removing a restrictive gate. Unlimited warping alone can create visibly impossible legs. A small, useful repertoire with generous but bounded entry ranges is a better prototype than either extreme.

The expensive failure mode in the brief is a chain of individually plausible requirements. The current plant must release, the other foot must support, the pelvis must transfer, a swing must clear the leg, and only then can contact begin. The compounded wait may be far larger

than any one requirement suggests. Those dependencies should be retained only where they prevent a gameplay impossibility or a conspicuous visual failure.

4.4 Touch cadence is an opportunity schedule

REASONABLE INFERENCE. A cadence determines how soon the ball can receive another purposeful change. If direction requests arrive uniformly through an unmodifiable cycle of length T, the average wait until its next contact is T/2 and the worst wait approaches T. This is an illustrative scheduling model, not a measured FC model.

RECOMMENDATION. Let context choose touch timing and size together. Open running can permit a larger touch and a longer interval. Tight carrying should retain a nearer next opportunity. A large new request should examine whether a shorter, reachable contact can replace the next ordinary touch.

Bringing a contact forward is conditional. The ball must be reachable and the authoritative foot proxy must have a continuous, legal approach. The system cannot simply add an invisible impulse whenever input changes. An early contact should replace the superseded planned touch, preventing a second stale impulse later in the cycle.

Do not force every root correction to have a ball contact. A small correction may change the player's approach to the existing ball path. Equally, do not produce decorative contacts merely to inflate responsiveness metrics.

4.5 Effort Touch is a useful middle layer

OFFICIALLY CONFIRMED HISTORICALLY. FC 24's manual Effort Dribble Touch uses R1 plus right stick or RB plus right stick. Flicking versus holding changes knock distance; it can be requested during carrying or as a first touch. That input mechanic predates the later contextual use discussed in section 2.1. [S13]

RECOMMENDATION. Give Touchline an urgent contact family between a routine carry and a named skill. Its purpose is to find an earlier useful redirect through a shorter approach, alternative foot or reaching motion. It needs no theatrical preparation and should exit promptly into ordinary locomotion.

Such a family can visually explain an aggressive reach or abbreviated step. It must still satisfy contact, reach and impulse constraints. The assertion that EA literally pulls its next scheduled contact forward is an unverified hypothesis; the proposed Touchline scheduler is a way to achieve the useful effect while preserving the requested ball architecture.

4.6 Braking is as important as turn animation

RECOMMENDATION. Give the simulation a deliberate braking response. An animation that faces the new direction cannot make an authoritative root stop travelling in the old direction. If old momentum is retained too long, the character feels like a vehicle even when the pose reacts quickly.

A football controller need not impose a vehicle's non-slipping forward axis. It can reduce the old velocity component, allow a small lateral step and build the new component. Retain continuous position and bounded velocity changes, but tune those bounds for gameplay. Use current speed to distinguish a small walking correction from a sprint cut.

At low speed, permit a compact pivot or shuffle. At high speed, show and simulate meaningful braking before the new exit. High Agility can shorten that transition; Acceleration still governs how quickly speed returns. No evidence here quantifies how far FC exceeds human acceleration or friction limits, so no percentage of sacrificed realism is defensible.

4.7 Rapid analog input should express current intent

RECOMMENDATION. Treat ordinary analog movement as a current desired state. Maintain the latest accepted direction and magnitude, rather than a first-in-first-out list of turns. Preserve discrete commands such as an explicitly requested skill separately.

For 0° → 90° → 0° → −90°, accept each change promptly and replace superseded, uncommitted contact plans. If input changes during a protected final approach, the already-committed contact may still complete if it remains physically valid; record that bounded commitment separately from a stale queued action. If its contact has executed, retain that physical result and recover toward the latest request. Never play an obsolete 90° turn after the stick has moved on merely to honour the old queue.

This does not require aggressive filtering. Use a small neutral deadzone and threshold hysteresis to prevent noise; avoid an indiscriminate smoothing window that erases deliberate reversals. Replanning contact intent and restarting a base animation are different operations. Many input updates should leave the current clip running with adjusted targets.

The exact EA policy for rapid request replacement remains unknown. The recommended policy follows from Touchline's stated control goal and the impossibility of completing arbitrarily frequent full turns with a free ball and bounded reach.

4.8 Lessons from other systems

OFFICIALLY CONFIRMED IN OTHER SOFTWARE. Konami's eFootball 6.0.0 notes address unnecessary dribbling motions, stronger-foot selection, backward turns, and inconsistent speeds on exit. They also reduce excessive dash speed for defensive balance. The useful lesson is to improve contextual selection and predictable exits, not merely increase playback speed. [S18]

Epic documents orientation and stride warping to fit a pose to movement, and motion warping to fit root motion to interaction targets. These are available design techniques, not evidence that FC uses Unreal or these exact nodes. Touchline should adapt the presentation to its simulation-approved path rather than letting a visual warp move gameplay colliders. [S19, S20]

EA's published Goal Driven Animation patent describes producing intermediate poses using current pose, target pose and trajectory information. It provides a relevant example of goal-conditioned presentation. A patent establishes a disclosed concept, not shipment in FC 26, and it supplies no reason to add machine learning to the smallest Touchline prototype. [S17]

5 What Touchline has likely been doing wrong

RECOMMENDATION BASED ON THE DESCRIBED SYMPTOMS. These are hypotheses to test against PLAYTEST-1.2, not claims about unseen source code. [B]

First, the body planner may have become a permission system. If a desired touch must wait until every support and clearance condition is ideal, presentation constraints have effectively gained authority over input timing. Ball outcomes can remain technically simulation-owned while animation eligibility still makes the game feel unresponsive.

Second, retention alone may be rewarding the wrong behaviour. A controller can keep the ball safely by moving slowly, taking a long curve, or ignoring difficult direction changes. The 99 reference needs simultaneous requirements for input fidelity, useful progression and control. Excellent possession statistics do not prove that the requested football action occurred.

Third, ordinary analog corrections may have been mapped onto actions that contain too much choreography. A Cruyff turn or Ronaldo chop has a recognisable preparation, contact sequence and exit. Reusing that complete sequence for every direction change adds commitments that the input did not request. A generic inside cut can resemble part of a named action without inheriting its full state machine.

Fourth, multiple independent correction systems may be fighting each other. Root smoothing preserves old velocity; a support solver preserves a plant; the swing planner chases a changing ball target; pelvis correction tries to make the legs fit. Each local solution can look defensible while their combined result jitters, crosses legs or repeatedly abandons steps.

Fifth, whole-cycle commitment may be larger than the meaningful event. The body should not remain unavailable for an entire recovery merely because a contact needed a short stable approach. The new design should protect contact continuity while allowing the rest of the character to respond.

Finally, fine-step anatomical scoring may be misaligned with the camera and the product. A small support-foot discrepancy visible only when paused may be acceptable. A delayed exit, repeated pelvis oscillation or misleading ball contact is apparent during normal play. The prototype must judge those at gameplay speed.

6 Recommended Touchline architecture

6.1 Define authority by gameplay consequence

RECOMMENDATION. Keep all decisions that can change possession, space, contact opportunity or collision outcomes in simulation. Put their exact skeletal expression in presentation.

Quantity	Authority	Rule
Accepted direction and magnitude	Simulation	Timestamped intent, reproducible from the input trace.
Root path and velocity	Simulation	Own braking, acceleration, collision sweeps and positional continuity.
Gameplay facing	Simulation	Own orientation used by shielding, reach, tackling or action eligibility.
Contact time and point	Simulation	Planned conditionally; executed only against the actual ball.
Outgoing ball state	Simulation	Integrate the legal contact impulse and subsequent physics.
Contact foot and gameplay technique	Simulation	Own any choice that changes timing, quality, exposure or interaction.
Gameplay body or foot proxies	Simulation	Own their continuous paths whenever they can collide or exert force.
Exact visual swing and support plants	Presentation	Adapt freely within contact and readability limits.
Visual stride phase and pelvis motion	Presentation	Fit the authoritative motion; cannot delay it.
Torso, head, arms and transition pose	Presentation	Communicate accepted intent with bounded offsets.
Foot IK and base clip choice	Presentation	Explain the contact; never create, cancel or postpone it.

The key refinement is the distinction between a **gameplay contact effector** and a **rendered foot**. The effector can be a simple, continuous foot-sized proxy with generous but finite reach and speed. It provides the physical witness for contact. The skeleton follows it near impact while taking substantial liberties elsewhere.

This avoids two extremes: deriving gameplay from the full rendered skeleton, or allowing a point beside the ball to appear and deliver an unexplained impulse. If a leg or body region participates in tackling or shielding collisions, its simplified collision proxy and motion also belong to simulation.

6.2 Keep a small authoritative state

The simulation needs the root transform and velocity, gameplay facing, current ball state, latest input, current contact plan, active-foot readiness, previous executed contacts and relevant player attributes. Include enough effector state to prevent instant relocation, implausibly rapid alternating contacts or unlimited impulse spam.

It does not need a solved anatomical support polygon, complete joint torque history or a mandatory exact pelvis transfer for routine carrying. A compact gameplay state such as

ordinary carry, braking, urgent reach or collision recovery can constrain capability without becoming a full biomechanics model.

Keep gameplay action capabilities in versioned simulation data. Do not use the currently rendered foot phase, loaded clip inventory, animation level of detail or IK convergence to decide whether a touch is legal. Asset validation should find presentation gaps during development; the renderer must not become a runtime veto over simulation.

6.3 Process input and update movement immediately

On each fixed simulation tick, accept the input samples assigned to that tick and update the desired direction and speed. Use deterministic resampling when render and input rates differ. Deadzone handling should preserve intentional low-magnitude movement.

Update braking and desired root acceleration from the latest intent. Evaluate movement in relation to the current velocity and ball position rather than forcing movement heading and body facing to coincide. A player can shuffle, open the hips or travel briefly sideways while preparing a new line.

For the smallest prototype, use the existing simulation rate. There is no need to assume that FC runs at the same rate or to change Touchline's rate to match an unverified number. If 60 Hz is used for an experimental timing budget, label its tick duration as 16.7 ms and record the actual render cadence separately.

6.4 Resolve a conditional contact opportunity

Predict the physical ball forward under the existing solver. Evaluate a small set of contact families against root motion, approach direction, urgency, active-foot readiness, legal reach and obstacles. Prefer a useful, timely and controllable next state, with a secondary preference for continuity and the player's stronger foot.

The resolver should choose a provisional contact time, foot, point and allowable impulse region. Its target outgoing state can reflect the intended next movement line, but the final state must result from a legal contact. Account for ball speed, height and spin, and reduce root speed when necessary to retain a controllable approach.

Replan only the unexecuted future from the actual current state. If input changes, bring the next contact forward when feasible, select another foot, change its direction, or cancel it. Never rewrite an already executed impulse. If a root collision changes the approach, revalidate the contact plan immediately.

6.5 Use a narrow and explicit commitment

RECOMMENDATION. Distinguish planning, commitment and execution:

1. **Provisional approach.** Direction, timing, foot and family can be revised within capability bounds. Presentation follows the latest plan.
2. **Committed approach.** A brief final segment protects the contact effector's continuity. New intent can still affect preparation for the next action and any independently legal root response.
3. **Executed event.** The actual incoming ball state, contact point, participant and impulse are recorded. The event is immutable within the replay timeline.

4. **Recovery.** Return quickly to ordinary control. Do not wait for decorative completion of a clip.

For an initial 60 Hz experiment, a two-to-three-tick commitment budget is a parameter worth testing, not a universal rule. Some approaches require longer; some small adjustments can stay flexible later. Preserve the contacting leg's continuity rather than freezing the entire body.

Commitment is never a reservation of the ball. A defender, wall, deflection or changed ball state can invalidate the opportunity. Use swept collision checks and deterministic time-of-impact ordering against the actual ball. If an opponent reaches it first, the scheduled dribble touch must not receive priority simply because it was planned earlier.

6.6 Send presentation a descriptive contract

The action/context description should provide the authoritative root trajectory, gameplay facing, action family, urgency, contact foot, provisional or committed status, anticipated contact time and point, and subsequent executed-contact records. A short lookahead helps the renderer prepare; it does not grant the renderer future authority.

Choose authored motion using current speed, turn demand, foot, ball-relative location and visual compatibility. Treat requested speed as an exit objective. A sprint arrival should not be made to fit a walking pivot solely because both end 90 degrees from the original direction.

Then retime the useful portion of the motion, warp its stride and pose, and use IK to clean up the visible contact. Distribute correction over the available approach. Do not reserve all the error for a last-frame foot snap.

Use a modest preference for retaining the current base motion when two choices are otherwise suitable. Small updates often need parameter changes, not branching. If a new base motion is necessary, use a short compatible transition that preserves pose continuity and, where practical, motion continuity.

6.7 Bound visual root adjustments

A mesh offset may improve contact presentation, but it must have a spatial limit, a bounded change rate and a settling rule. It must not alter the authoritative player location, collision reach or shielding geometry.

Judge the limit by whether the visible player still represents the space the simulation says they occupy. A few centimetres of correction might be unobjectionable at one camera and misleading in a close view. Set the budget experimentally rather than borrowing an invented EA distance.

If a larger root-path change is needed for gameplay, ask the simulation movement controller to make it as an explicit legal change. Calling a gameplay displacement a visual warp does not make it presentation-only.

6.8 Build a generic contact vocabulary first

Start with forward carry, compact inside redirect, compact outside redirect, sole check or stop, across-body gather, urgent reach, and a braking pivot with a short exit. Include appropriate left/right and incoming-speed coverage. These are functional families; they need not be named tricks in the user interface.

A named skill belongs in the gameplay action catalogue when its deception, ball exposure, extra contacts or movement path matter. Ordinary left-stick input should not automatically request a Ronaldo chop, Cruyff or drag-back merely because a target angle happens to resemble its exit.

Keep preferred-foot behaviour as a scored preference subject to opportunity and quality. A 99 Dribbling value should not silently supply a perfect weak foot. Record independent weak-foot and physical attributes when testing the reference player.

7 What should deliberately cheat

RECOMMENDATION. Spend visual freedom on removing dead time and preserving the shape of the football action.

Presentation freedom	Intended benefit	Boundary
Abbreviated support transfer	Start the useful response sooner.	Preserve readable weight shift; no full-body freeze to preserve one plant.
Shortened or repositioned support step	Fit late direction changes.	Avoid conspicuous skating or continuous foot dragging.
Small plant rotation	Permit compact pivots.	Prefer releasing the plant to twisting the leg visibly.
Stride and timing adjustment	Fit current speed and contact timing.	Do not move the ball event to fit the clip.
Pelvis, torso and limb pose warping	Maintain a coherent silhouette.	Keep distortion and visible collider mismatch bounded.
Early recovery exit	Return to useful control promptly.	Preserve the contact witness and executed result.
Limited non-contact self-overlap	Avoid overconstraining fleeting transitions.	Do not let a visual limb imply an impossible ball or opponent interaction.

Use authentic-looking reaching and imperfect recovery poses when they explain a difficult action. Perfect balance should not be required for every controlled touch. Smoothness should come from a coherent whole-body gesture, not from forcing every joint to satisfy an independent ideal.

Aggressive braking is a separate category. A braking animation is presentation, but the corresponding change in root velocity is gameplay. Tune it openly in simulation. The same

applies to acceleration after a cut: a faster-looking stride must not conceal an unchanged, sluggish authoritative root.

Do not interpret tolerated imperfection as permission for obvious errors. Frequent foot skating, rubber limbs, repeated stance resets and false contacts damage readability. The proposed policy is to allow small, bounded discrepancies while preserving the events the player actually uses to understand the game.

8 What must remain physically and authoritatively correct

RECOMMENDATION. Preserve the following invariants:

1. **One ball state.** Its position, velocity, spin and collisions belong to simulation. No foot parenting, visual substitute ball, post-hoc relocation or unexplained attraction toward the carrier.
2. **A physical cause for each touch.** A continuous authoritative effector must reach the actual ball. The event applies a bounded action impulse; presentation does not decide success.
3. **World interactions remain effective.** Opponents, terrain and other collisions can change or remove a planned opportunity. There is no possession guarantee attached to an animation.
4. **No retroactive correction.** A new request cannot change an earlier touch or repair its history. A missed committed touch cannot reappear later as a stale impulse.
5. **Continuous root and effector motion.** Generous gameplay capabilities still need finite reach, finite travel and defined collision rules. Replanning must not spawn a new foot beside the ball.
6. **Deterministic outcomes.** The same simulation inputs and starting state produce the same authoritative event trace, independent of render rate, visual variant, blend duration or IK iteration count.
7. **Clear collision ownership.** Gameplay-facing and proxy geometry determine interaction. Purely visual offsets cannot extend tackle reach or change shielding.

Between touches, the ball can respond to gravity, spin, drag, friction and genuine contacts. It must not turn toward a newly requested line solely because the next animation wants it there. The simulation may choose an outgoing impulse at a valid touch that makes the following action easier; that is intentional control with a visible cause.

The authority rule should be tested with the renderer disabled. Re-enable it with a different visual stride phase or motion variant and require the same ball and player-state hashes. A contact log alone is insufficient if hidden root or collision changes can still leak into gameplay.

9 Proposed 99 99 99 reference behaviour

9.1 Define the guarantee within a controllable envelope

RECOMMENDATION. Ball Control 99, Dribbling 99 and Agility 99 should mean very reliable technical execution, excellent adaptation of the next touch, and rapid reorientation within the player's remaining physical and technical capabilities.

Attribute	Reference behaviour
Ball Control 99	Minimal avoidable touch error in a reachable, unopposed situation; accurate receiving and carrying targets.
Dribbling 99	Efficient contact choice, tight correction opportunities, useful retained speed and short recovery.
Agility 99	Prompt braking and reorientation, economical footwork, rapid transition onto a different line.

The combination should not imply 99 Sprint Speed, Acceleration, Balance, Strength, Reactions or weak-foot ability. Hold those inputs constant in an A/B test and record them. A high-agility player with modest acceleration can turn neatly but should not regain sprint speed like a faster player.

Near-zero unforced loss is appropriate inside an explicit technical-control envelope: ordinary ball heights and speeds, reachable separation, available legal contact, and no intervening opponent. It cannot mean automatic possession through a defender or an instant reversal of a ball already knocked beyond reach.

Under rapid input, the elite player should preserve control by updating the next useful opportunity and, where appropriate, shortening the next touch or braking. It should not quietly preserve retention by ignoring the intended new line. For genuinely late input after a large sprint touch, immediate preparation followed by the earliest feasible recovery is the correct response.

9.2 Initial timing budgets

All numbers in this subsection are **proposed Touchline prototype targets**. They are not measured FC values, human physiological limits or promises for every game context. Use the actual project tick rate; 60 Hz below is an example.

Measure	Initial target or rule
Accepted input to authoritative acknowledgement	Within the next simulation tick; 16.7 ms at 60 Hz.
Accepted input to first directional visual response	Aim for p95 at or below 50 ms in the controlled local test.
Relevant contact in a predeclared reachable close-ball turn	Start by testing a p95 budget of 150 ms; report failed or unavailable cases.

Measure	Initial target or rule
Final contact commitment	Explore 2-3 ticks at 60 Hz, extending only for a concrete continuity need.
Stale analog actions	Zero execution of superseded uncommitted plans; log valid already-committed completions separately.
Technical loss in the small deterministic suite	Zero unforced failures in eligible cases, with exclusions reported.

Do not force sprint contact latency to match close-control latency. Its ball separation and effector availability may require a longer wait. Meaningful new-line time should improve materially over PLAYTEST-1.2 without being achieved by a low-speed spin in place.

Touch cadence and separation should initially be distributions, not universal targets. Close carrying should generally offer shorter opportunities than an exposed sprint carry; an urgent contact is an exception that replaces a plan, not an extra repeated correction. Tune these jointly with movement speed and normal-camera appearance.

9.3 Metrics that resist gaming

Metric	Operational definition and safeguard
First visible response	First directionally relevant pose or braking cue above a preregistered perceptual threshold. Exclude indicators and unrelated twitches.
First authoritative response	First accepted-intent-driven change to braking, velocity or gameplay facing. Separate from purely visual movement.
First relevant contact	First executed contact that usefully reduces error to the still-current request. An irrelevant tap does not qualify.
Meaningful heading change	Time to a chosen angle of actual velocity redirection. Do not calculate heading for near-zero speed.
New movement line	Velocity within a chosen tolerance of the target and above a minimum useful speed for a short dwell.
Lock time	Duration in which a feasible new request cannot alter the current or next action. Report its cause.
Branch availability	Fraction of sampled states with a useful bounded adaptation, plus longest unavailable run. Do not maximise branch count.
Ball separation	Raw root-ground-point to ball-centre distance distribution; also report relation to the shared reach envelope.
Control loss	Separate technical error, intentional exposure, unavailable contact and legitimate opponent action.
Contacts per second	Count executed events by context; report burstiness and redundant touches. More is not automatically better.

Metric	Operational definition and safeguard
Exit acceleration	Time from the turn to a specified useful speed on the new line, with Acceleration held constant.
Input fidelity	Error to still-current intent and count of stale actions; prevents retention through ignored input.
Visual readability	Blinded normal-speed judgement of the action, exit and ball independence. Slow motion diagnoses causes.

Use median, p95 and the worst observed result by speed and contact phase. Means can hide the occasional long lock that dominates player frustration. Keep hardware stick-to-display measurements separate from logs beginning at simulation input acceptance.

For a concrete initial new-line metric, one candidate is heading within 15 degrees, speed at least 40 percent of requested speed, maintained for 100 ms. These thresholds should be fixed before A/B comparison and supplemented by travel distance toward the request. Otherwise a controller can win through slow pivots that do not deliver useful movement.

Do not allow each controller to define its own eligible subset. Use a shared reach/capability envelope and record every request as current and reachable, superseded, physically unavailable, or externally disrupted. Report all categories, including the cases where no timely contact occurred.

Give each sustained request a fixed evaluation deadline shared by A and B. A still-current request that does not produce the required contact or useful new line by that deadline is a failure and a right-censored latency observation, not a dropped sample. Report completion rates alongside latency; do not claim a faster p95 if failed completions make that percentile unobservable.

10 Prototype against PLAYTEST-1.2

10.1 Scope and variants

RECOMMENDATION. Use one existing elite player, the existing physical ball, a flat pitch and one common camera. Freeze PLAYTEST-1.2 as A. Build B only when implementation is separately requested, with responsive root braking, a revisable next contact, a small generic touch repertoire and adaptable presentation. There is no need for a new motion-capture dataset, machine-learning model or whole-body dynamics project to test the hypothesis.

Keep initial states, input traces, attributes, pitch conditions, opponent commands and collision laws fixed across A and B. Once root paths and touch schedules differ, collision outcomes may legitimately differ. Forcing identical collisions would invalidate the behavioural comparison.

The first A/B tests the package. It cannot identify the isolated contribution of every component. That is acceptable for the smallest experiment; avoid a large factorial project before establishing that the direction is useful.

10.2 Twenty four paired baseline sequences

Run three input families at jog and sprint, each at four deterministic offsets through the authoritative incoming contact cycle. That produces 24 paired sequences:

1. A held 90-degree turn, sustained long enough to establish a new line.
2. A reversal placed around the proposed contact commitment boundary, with before/after variants distributed across the four offsets.
3. The rapid 0° → 90° → 0° → −90° request, followed by a held final direction.

Use the same scripted timing in A and B. Do not move the requested reversal separately to each implementation's preferred contact phase after seeing its result. Report the actual commitment relationship in each trace.

Add a short manual exploration of slow small corrections, stick release, full rotation and sprint-to-cut. Mirror the cases where preferred-foot asymmetry is material. These are diagnostic checks, not extra evidence disguised as a larger preregistered sample.

10.3 One physical contest gate

Add a defender whose authoritative movement reaches the ball before a scheduled dribble contact. The earlier collision must be respected; the attacker must not receive a delayed correction that restores the old plan. Include a control case where the defender arrives after the valid dribble contact.

This gate distinguishes a physical contact scheduler from a possession script. It is necessary even when the initial problem concerns unopposed control, because an apparently excellent isolated dribbler can hide an invalid ball guarantee.

10.4 One inexpensive visual isolation check

Replay the same already executed A trace through the old and proposed presentation on motion both can support. That isolates the visual contribution without changing ball results. If B's trace cannot be represented by the old body planner, record that incompatibility rather than silently relaxing A.

Separately vary only presentation phase or visual variant during B's simulation replay. The gameplay hashes must remain identical. Changing authoritative foot readiness is a different experiment and may legitimately change contact timing.

10.5 Decision criteria

Advance B if it produces earlier useful response, materially reduces long waits, preserves technical control in eligible cases, and is preferred for smoothness and readability in blinded normal-speed comparisons. It must also pass deterministic replay and the defender gate.

Before tuning, declare useful-new-line delay as the primary outcome, identify the affected-case subset from the frozen baseline, and fix its shared eligibility rules, dwell threshold and evaluation deadline. As an initial decision threshold, seek at least a 25 percent reduction in that subset's p95 new-line delay. Require no increase in missed completions or technical-control failures, and no meaningful regression in relevant-contact latency or the other input families. Do not switch to a more favourable metric or subset after seeing B. This

is a proposed discriminator, not a scientifically established minimum perceptible difference. Show individual traces because 24 cases provide a coarse p95 estimate.

Reject a superficial win if retention improves by ignoring input, if pose response improves while useful ball/root response does not, or if frequent corrective touches make the motion look frantic. Reject any ball correction lacking a legitimate contact cause.

Zero failures in this small suite does not establish a population-wide near-zero loss rate. A broader phase, speed and input-duration sweep becomes useful after B wins the basic comparison. The first prototype's purpose is to test whether removing detailed body mechanics from the response gate produces a worthwhile improvement under the stated authority constraints.

11 Sources

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